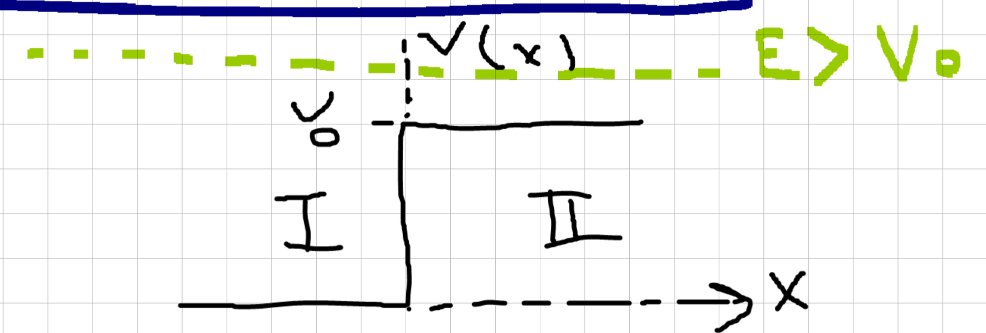
Quantum Mekaniği I Bölüm 5

Tekrar

$H = -\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)$ 1 Boyutta Hamiltonyen

$HU = EU$

1. Potansiyel Basamağı



$V(x) = \begin{cases} V_0 & x > 0 \\ 0 & x \leq 0 \end{cases}$

$HU = EU \Rightarrow -\frac{\hbar^2}{2m}\frac{d^2U}{dx^2} = EU \Rightarrow \frac{d^2U}{dx^2} + \frac{2mE}{\hbar^2}U$ $\quad \frac{2m|E|}{\hbar^2} = k^2 \Rightarrow D^2 + k^2 = 0$

$D = \mp ik$

$U(x) = \begin{cases} e^{ikx} & \cos kx \\ e^{-ikx} & \sin kx \end{cases}$

%100 = 1 $\quad$ Reflection $\quad$ ikx sağa gitme, −ikx sol

$\Rightarrow U_I(x) = 6e^{ikx} + Re^{-ikx}$

$U_{II}(x)$ için ; $-\frac{\hbar^2}{2m}\frac{d^2U_{II}(x)}{dx^2} + V(x)U_{II}(x) = EU_{II}(x) \Rightarrow -\frac{\hbar^2}{2m}\frac{d^2U_{II}(x)}{dx^2} = (E-V)U_{II}(x) \Rightarrow \frac{d^2U_{II}(x)}{dx^2} + \underbrace{\frac{2m(|E|-V)}{\hbar^2}}_{q}U_{II}(x) = 0$

$D^2 = \mp iq \qquad \underbrace{Ae^{iqx}}_{J} + \underbrace{Be^{-iqx}}_{0}$

sağa gelildiğinden

Transportion

$\Rightarrow Te^{iqx}$

$U_I = e^{ikx} - Re^{-ikx} \qquad U_{II} = Te^{iqx} \Rightarrow$ $\quad$ Continuityden; $\quad U_I = U_{II}$, $\quad U_I' = U_{II}'$

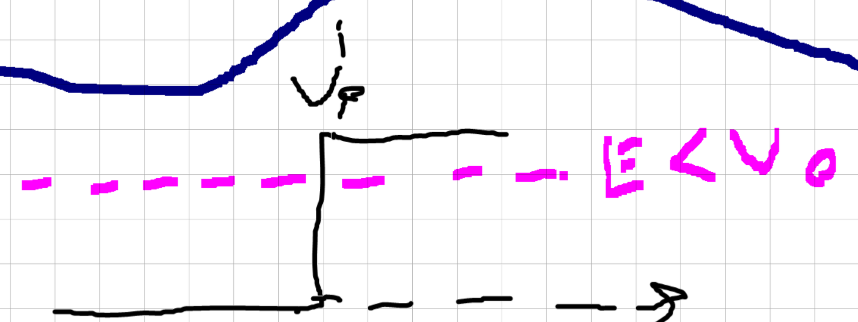
$\Rightarrow \begin{cases} 1e^{ikx} + Re^{-ikx} = Te^{iqx} \\ ik(e^{ikx} - Re^{-ikx}) = iqTe^{iqx} \Rightarrow \end{cases} \quad \begin{matrix} 1+R = T \\ k(1-R) = qT \end{matrix} \Rightarrow \begin{matrix} 1+R = T \\ 1-R = \frac{q}{k}T \end{matrix}$

$+$

$2 = \frac{k+q}{k}T$

$\Rightarrow \boxed{T = 2k/k+q} \Rightarrow 1+R = \frac{2k}{k+q}$

$\Rightarrow R = \frac{2k}{k+q} - 1 \Rightarrow \boxed{R = \frac{k-q}{k+q}}$



$U_I(x) = 1e^{ikx} + Re^{-ikx}$

U_{II} durumlar değişir $\quad$ Kuantum mek. Klasikten ayrılır budur

$-\frac{\hbar^2}{2m}\frac{d^2U_{II}(x)}{dx^2} + V(x)U_{II}(x) = EU_{II}(x) \qquad E < V_0$

$-\frac{\hbar^2}{2m}\frac{d^2U_{II}(x)}{dx^2} = (E-V)U_{II} \Rightarrow \qquad V > |E| \qquad |E| > V$

$\frac{d^2U_{II}}{dx^2} = -\frac{2m}{\hbar^2}(|E|-V)U_{II} \Rightarrow \frac{d^2U_{II}}{dx^2} + \frac{2m}{\hbar^2}(|E|-V) = 0$

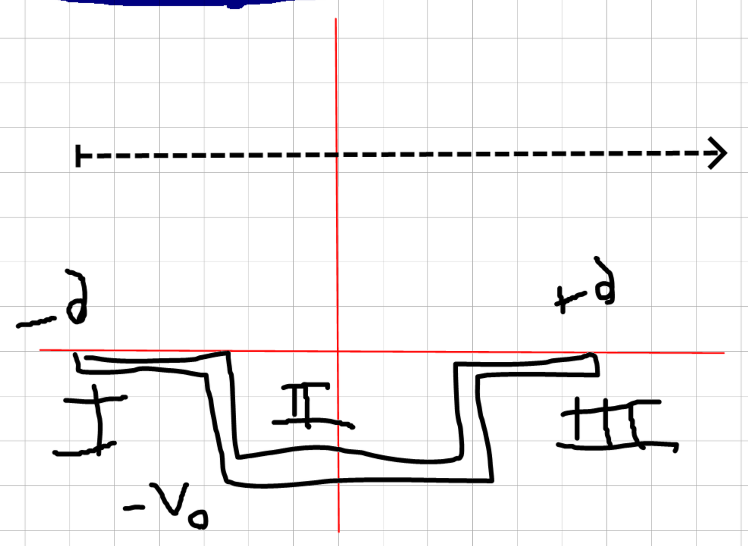
$\frac{d^2U_{II}}{dx^2} - \frac{2m}{\hbar^2}(V-E) = 0 \qquad D^2 - K^2 = 0$

$D = \mp K$

$T = \frac{2k}{k+K}$

$\frac{k-K}{k+K} = R$

2. Potansiyel Kuyusu



$E>0$ saçılma $E>V$ $HU=EU$

$$-\frac{\hbar^2}{2\mu}\frac{d^2U}{dx^2}+VU=EU$$

$U_{III}=U_I \Rightarrow V_0=0$ $\quad U_{II}:-V_0$

$$\Rightarrow -\frac{\hbar^2}{2\mu}\frac{d^2U_I}{dx^2}=EU_I \qquad -\frac{\hbar^2}{2\mu}\frac{d^2U_{III}}{dx^2}=EU_{III} \qquad -\frac{\hbar^2}{2\mu}\frac{d^2U_{II}}{dx^2}+VU_{II}=EU_{II}$$

$\Rightarrow$ I için $\frac{d^2U_I}{dx^2}+\frac{2\mu E}{\hbar^2}U_I=0 \Rightarrow D^2+k^2=0$

$D=\mp ik \qquad 1e^{ikx}+Re^{-ikx}=U_I \;;\; U_{III}=Te^{ikx}=$ III için

II için $\Rightarrow -\frac{\hbar^2}{2\mu}\frac{d^2U_{II}}{dx^2}-VU_{II}=EU_{II}$

$$\Rightarrow -\frac{\hbar^2}{2\mu}\frac{d^2U_{II}}{dx^2}=(E+V)U_{II} \Rightarrow \frac{d^2U_{II}}{dx^2}+\underbrace{\frac{2\mu(E+V)}{\hbar^2}}_{q}U_{II}=0 \Rightarrow D^2+q^2=0 \qquad D_{1,2}=\mp iq$$

$$Ae^{iqx}+Be^{-iqx}$$

$U_I=U_{II} \qquad U_{II}=U_{III}$

$U_I'=U_{II}' \qquad U_{II}'=U_{III}'$

$\Rightarrow$ ✱ $e^{ikx}+Re^{-ikx}=Ae^{iqx}+Be^{-iqx}$; $Ae^{iqx}+Be^{-iqx}=Te^{ikx}$

$ik(e^{ikx}-Re^{-ikx})=iq(Ae^{iqx}-Be^{-iqx})$; $iq(Ae^{iqx}-Be^{-iqx})=ikTe^{ikx}$

$x=-a \qquad\qquad x=a$

$e^{-ika}+Re^{+ika}=Ae^{-iqa}+Be^{+iqa} \qquad Ae^{iqa}+Be^{-iqa}=Te^{ika}$

$ik(Re^{-ika}-Re^{ika})=iq(Ae^{-iqa}-Be^{iqa}) \qquad iq(Ae^{iqa}-Be^{-iqa})=ikTe^{ika}$

$$R=\frac{ie^{-ika}(q^2-k^2)\sin 2qa}{2kq\cos 2qa-i(q^2+k^2)\sin 2qa}$$

$$T=\frac{e^{-i2ka}2kq}{2kq\cos 2qa-i(q^2+k^2)\sin 2qa}$$

Not R ve T değerleri Sonraki Pdf te